Pneumothorax

Review Video: https://coreultrasound.com/pneumothorax/

Knobology/Preparation
- Probe: curvilinear, linear (best image quality)
- Depth: **10cm** and then center pleural line
- Position: supine
- Present: lung/abdomen
- Orientation: longitudinal

Landmarks
- **External:** most anterior part of lung in mid-clavicular line of a supine patient
- **Internal:** ribs/rib shadows
- **AOI:** pleural line

What to assess for:
- **Lung Sliding:** visceral and parietal pleura moving against each other with respiration
  - Looks like ants sliding on a log
- **Comet tails:** short white line arising from the pleura line caused by reverberation of the US beam between visceral and parietal pleura
- **Lung Pulse:** cardiac pulsation transmitted to the pleural line in a poorly aerated lung (i.e atelectasis or main stem intubation)
- **Lung Point:**
  - **Physiological lung point -** sliding next to something else that is also sliding
    - Cardiac, liver, spleen, gastric
  - **True Lung Point (pathological) -** sliding next to no sliding (100% specific)

Steps:
1. Place probe in **longitudinal** plane at most anterior aspect of one hemithorax (usually around intercostal space 4-5)
2. Identify the **ribs/shadows** = internal landmark
3. Identify the area of interest: **echogenic pleural line** (representing the normal visceral and parietal pleural)
   - Pleural line is 0.5 -1.5cm far field of the ribs
   - Sweep this to optimize the crispness of the pleural line.
4. Evaluate the pleural line for pneumothorax, do this in 3 pleural spaces.
   - Should be evaluating for lung sliding, lung pulse and comet tails
5. **Lung point:** (most specific thing for diagnosing a pneumothorax)
   - If seen, assess to see if it is physiologic lung point
   - If lung sliding, comet tails and lung pulse are not visible in one or more intercostal spaces for at least 3 respiratory cycles AND the patient is stable, a lung point should be identified to declare a pneumothorax
     - To do so: start in anterior axillary line → slide to mid axillary line → slide to posterior axillary line, stopping if lung sliding/comet tails are seen at any point.

Approximate sizes of pneumothorax in supine patient:
- **Small =** lung point located anteriorly - i.e. between mid-clavicular and anterior axillary lines
- **Medium** = lung point located laterally (i.e. mid axillary line)
- **Large** = lung point located posteriorly (i.e posterior axillary line)

Troubleshooting:

- Sweep to ensure beam perpendicular to pleura
- Decrease depth to magnify pleura
- Decrease gain (may get wash out)
- Adjust probe frequency (higher frequency = better)
- Rotate probe to elongate pleura
- Change to linear probe
- Differentiating between physiologic from true lung point
    - Increase depth to appreciate the solid organ or lack thereof.

Determinate Negative Scan:

- **Lung sliding OR Lung pulse OR Comet Tails / B-lines** present in most anterior rib space.

Determinate Positive Scan:

- Absence of lung sliding, comet tails AND lung pulse in at least 1 rib space (make sure to observe over 3 resp cycles)
    - If stable: find lung pulse
    - If unstable: okay to declare pneumothorax without finding a lung pulse

Pitfalls:

- **Image interpretation**
    - **False Positives:** (no lung sliding will be present – ensure to look for lung pulse/comet tails)
        - R. mainstem intubation → look for comet tails/lung pulse in L. lung
        - Esophageal intubation in an apneic patient
        - Esophageal intubation in an apneic patient AND in cardiac arrest
        - Phrenic nerve palsy - isolated absence of lung sliding
        - Conditions in which the visceral and parietal pleura are adherent to each other
            - ARDS
            - Chronic pleurodesis
        - Pulmonary fibrosis: Should have comet tails/lung pulse
        - Large pulmonary infiltrates
        - Pleural effusion: no pleura visible
        - Severe COPD with bullae formation: no plera visible
        - Pulmonary contusions - comet tails should still be evident, but lung sliding will be MIA.
    - **Interpreting physiologic lung points as pathologic lung points**
        - Cardiac, splenic, gastric, liver
    - **False Negatives:**
        - Large pneumothorax (> 65%) no normal lung tissue touching an accessible part of the pleura in a supine patient
        - Misidentifying movement at the pleural line due to poor had control, as lung sliding/pulse
- **Image generation**
    - Sweeping too quickly
    - Not interrogating pleura at 90 degrees
    - Poor control / grip
    - Overgain
    - Pleura not centered / depth not appropriate
    - Not waiting 3 respiratory cycles
- **Clinical Integration**
    - Treating for PTX based on false positive
    - Treating a very small pneumothorax (remember, some don't need a chest tube!)

- Not treating based on a false negative scan or indeterminate scan, despite high clinical suspicion